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Laminar flame characteristics and chemical kinetics of 2-methyltetrahydrofuran and the effect of blending with isooctane

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A B S T R A C T

2-methyltetrahydrofuran (MTHF) has been considered as a potential biofuel candidate for the renewable feedstock and attractive properties. In this study, the spherically propagating flames of the MTHF/isooctane-air mixtures in different blending ratios were investigated at elevated temperatures and pressures over equivalence ratios of 0.8–1.5 in a constant volume chamber using high-speed photography technique. Laminar burning velocities were calculated through nonlinear method and correlated by mixing rules as a function of initial temperature, pressure, equivalence ratio, and blending ratio. To investigate the influence of the potential interaction between MTHF and isooctane, a detailed model was established by merging the models of the two fuels and validated against laminar burning velocity, ignition delay times and flame structure.

The laminar flame velocities of MTHF were observed to increase with initial temperature while the flame velocities decrease with increase in pressure. Kinetic analysis indicates that the major consumption pathways for MTHF are through H abstraction reactions at 2 and 5 sites. The H abstraction reactions at methyl group are less competitive. Unsaturated hydrocarbons and aldehydes are produced in high concentration and are the main stable intermediates of MTHF combustion. The laminar burning velocity is sensitive to the reactions of small radicals and some intermediates such as propylene and ethylene.

The laminar burning velocities of MTHF/isooctane blends increase with MTHF blending ratio. The influence of blending ratio on laminar burning velocity should attribute to the chemical factors, rather than thermal or diffusion factors. MTHF oxidation generates less proportion of propylene and higher fractions of H, OH than isooctane, reflecting positive chemical effects on laminar burning velocity as MTHF blended. According to the instability analysis, the Markstein length and critical radius shows similar flame instability among different blending ratios.

1. Introduction

As traditional fossil fuels are largely responsible for energy crisis and atmospheric degradations, there is an increasing attention to biofuels. 2-methyltetrahydrofuran (MTHF) is considered as a promising biofuel for its attractive properties. MTHF could be produced from agricultural and forestry wastes using furfural [1] and levulinic acid [2] as platform molecules. As an important component of P-series fuels which are a series of renewable, non-petroleum fuels providing significant emissions benefits, MTHF plays an important role in reducing fuel volatility and promoting environmental quality [3,4]. In "Tailor-Made Fuels from Biomass" project of RWTH Aachen University [5], MTHF was also selected as a promising candidate for alternative transportation fuels.

MTHF has been studied experimentally [5–11] to understand the engine performances and emissions. García et al. [5] studied fundamental spray characteristics related to fuel vaporization and fuel/air mixing for MTHF and reported extremely low soot emissions in a single-cylinder CI engine indicating that MTHF is very suitable for mixing-controlled combustion. MTHF shown higher enleanment potential and ignition prone than RON 95 fuel in a study on a direct injection single cylinder research engine [6]. Thewes et al. [7] compared the knock resistance and required compress ratios of MTHF with several other bio-fuel candidates on homogeneous engine combustion systems. They concluded pure MTHF is knock restricted and requires a low compression ratio as its low RON. Autoignition characteristics and derived cetane numbers (DCN) of pure MTHF was also investigated experimentally using motored-engine technique [8] and ignition quality tester [9].

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Fuel design provides an innovative and sustainable way for the development of internal combustion engines. As a bio-derived oxygenated fuel of low octane number, MTHF exhibited excellent performance through blending with other fuels in new concept combustion modes. Janssen et al. [10] reported that the blending of MTHF and dibutylether used in homogeneous low-temperature combustion is a potential way to realize ultra-low soot emissions. Heuser et al. [11] reported the low PM, HC, CO emissions and noise of the blend fuels of dibutylethe and MTHF on a single-cylinder diesel research engine.

Gasoline direct injection (GDI) engine is notorious for its much higher particulate emissions than traditional gasoline engine, and it is a great challenge to meet the increasingly stringent emission standards. As MTHF is known to reduce particulate emissions, the blending with gasoline are considered could reduce the particulate emission on GDI engines, and the optimal blending ratios will depend on the trade-off with the engine efficiency [12]. There are some studies about the performances and emissions of engines fueled with MTHF/gasoline mixtures. Rudolph et al. [13] reported the SI engine fueled with gasoline/MTHF blend containing 10% MTHF has comparable power outputs and CO, NO_x, and NMHC emissions to unleaded gasoline. Lucas et al. [14] reported the HC, CO and benzene emission decrease evidently in a field trial using fuel blend consisting of 60% MTHF and 40% gasoline in volume.

Many studies were carried out to explore the combustion characteristics of MTHF. Simmie [15] conducted theoretic study at CBS-QB3 and G3 level for bond dissociation energies (BDE) of MTHF, barrier heights, and reaction enthalpies of all possible H abstraction reactions by H atoms and methyl radicals. The ignition delay times of MTHF at low to high temperature were investigated in shock tubes and rapid compression machines [6,9,16–20]. The structures and intermediates of laminar premixed low-pressure MTHF flames were measured using gas chromatographs (GC) and mass spectrometry (MS) technique [21–23]. The laminar burning velocities of MTHF/air mixtures at atmosphere pressure were measured in flat flames at 298–398 K [21] and spherically propagating flames at 333–393 K [24]. Moreover, the reactivity of MTHF with OH [25] and HO_2 [26] were also studied. Based on experiments and calculations, pyrolysis and combustion chemistries of MTHF were developed [18,21,22].

Laminar burning characteristics at elevated pressures and temperatures which resemble to the conditions in internal combustion engines are essential for the design and optimization of combustion chamber as well as combustion simulation in engines. Although the above studies have significantly improved the understanding into the combustion chemistry of MTHF, the laminar burning characteristics at elevated pressures are still not studied yet. Besides, the excellent emission characteristics of MTHF blend fuels on engine studies suggest that it is necessary to study the combustion characteristics of MTHF blends.

In this paper, the spherically propagating flames of the mixtures of MTHF–air and MTHF/isooctane–air were explored in a constant volume chamber using high-speed photography technique. The experiment was performed at the temperatures of 373, 423 and 453 K, the pressures of 1, 2 and 4 bar, the equivalence ratios of 0.7–1.6 with MTHF blending ratios of 0%, 20%, 40%, 60% and 100%. The laminar burning velocity, Markstein length and flame instability were discussed. To understand the combustion chemistry of MTHF and the kinetic interaction with isooctane, a detailed model for MTHF/isooctane high-temperature oxidation was developed based on the work of Bruycker et al. [21] and Chaos et al. [27], and adopted to analyze the combustion process.

Table 1
Operating conditions for MTHF/isooctane–air mixtures in this experiment.

Temperature (K)	Pressure (bar)	Equivalence ratio	MTHF blending ratio (in mole)
373	1	0.7–1.5	0, 20%, 40%, 60%, 100%
423	1, 2, 4	0.7–1.5	0, 20%, 40%, 60%, 100%
453	1	0.8–1.5	0, 20%, 40%, 60%, 100%

2. Experimental and computational methods

2.1. Experiment setup

The present facility has been specified in detail in previous works [28] and will be described here briefly. The experimental apparatus consists of a cylinder chamber of 0.0055 m^3 in volume, a gas supply system, a schlieren optical system, an ignition system and a timing control system. Heating tapes wrapped around the apparatus aid in modifying the initial temperature. Pressure transducer with the deviation of 1% and thermocouples with the accuracy of ± 2 K are installed inside the chamber to monitor the initial temperature and pressure. Each component of the fuel/air mixtures is induced into the chamber individually with the amount controlled by Dalton's law. Before every ignition, the mixture was mixed by molecular diffusion for at least 5 min to guarantee homogeneousness. Two horizontal tungsten electrodes with the radius of 0.5 mm are utilized to ignite the mixtures. After each ignition, fresh air is induced to scavenge the residual gas for at least 3 times. Two circular quartz windows of 80 mm in diameter on each side of the chamber supply the optical path for the schlieren system. A high speed camera (Phantom V611) is utilized to capture the ignition and flame propagation at a sample rate of 10 kHz and a resolution of 752 × 752 pixels. Raw data of the combustion characteristics would be obtained from the schlieren images.

The MTHF and isooctane used for current experiment are from Aladdin Industrial Corporation with the purity of 99%. N_2 and O_2 of ultra-high purity (99.999%) were induced in the ratio of 3.76 to simulate the composition of dry air. The MTHF blending ratios of 20%, 40% and 60% in mole are marked as M20, M40 and M60 in this study. The experiment conditions are listed in Table 1. For each operating condition, the experiment repeated at least 3 times to reduce the random uncertainty.

2.2. Data processing

The flame radius could be obtained from the schlieren pictures of flame propagation. In order to avoid the effect of ignition and cylindrical confinement on the flame propagation, the data with flame radius of 8–22 mm was selected for subsequent processing.

The stretched flame speed (S_b) could be derived from the relation between flame radius (r_f) and time (t):

$$S_b = dr_f/dt \tag{1}$$

The stretch rate, κ, of the spherical flame surface (A_f) could be deduced.

$$\kappa = \frac{\mathrm{d}A_f}{A_f \cdot dt} = 2S_b/r_f \tag{2}$$

The flame speeds of most fuels can be considered to vary linearly with the stretch rate and the unstretched flame speeds could be extrapolated according to $S_b = S_b^0 - L_b \cdot \kappa$, where S_b^0 is the unstretched flame speed, L_b is the Markstein length. However, strong nonlinearity was found for heavy fuels at fuel-rich conditions, casting considerable uncertainty on the conventional method of linearly extrapolating. As a consequence, the nonlinear model [29] was employed for the calculation of unstretched flame speed in this study.

$$\ln(S_b) = \ln\left(S_b^0\right) - S_b^0 L_b \cdot 2/\left(r_f S_b\right) \quad (3)$$

The following relations could be deduced by mass conservation.

$$S = S_b^0 \cdot \rho_b/\rho_u = S_b^0/\sigma \quad (4)$$

where S is the laminar burning velocity, σ is the density ratio, ρ_b and ρ_u are the densities of burned and unburned mixtures, respectively.

The effective Lewis number [30], a weighted average of the individual Lewis numbers of the fuel and oxidizer, is evaluated as:

$$Le_{eff} = \begin{cases} \frac{Le_o+(1-\tilde{\phi})Le_F}{2-\tilde{\phi}} & \phi < 1 \\ \frac{Le_F+(1+\tilde{\phi})Le_O}{2+\tilde{\phi}} & \phi > 1 \end{cases} \quad (5)$$

$$\tilde{\phi} = Ze(\phi - 1) \quad (6)$$

$$Ze = Ea(T_{ad} - T_u)/RT_{ad}^2 \quad (7)$$

where Le_{eff} is the effective Lewis number, Le_F and Le_O is the Lewis numbers of the fuel and oxidizer [30], Ze is Zeldovich number, E_a is the active energy, T_{ad} and T_u is the adiabatic flame temperature and initial temperature, R is the gas constant, ϕ is the equivalence ratio. The effective Lewis numbers of the blend fuels are derived as the average value of pure fuels weighted by mole fractions.

The flame thickness is identified as the ratio of thermal diffusivity to laminar burning velocities.

$$\delta = \alpha/S \quad (8)$$

where δ is the flame thickness and α is the thermal diffusivity.

2.3. Simulation method

To investigate the chemical oxidation of MTHF/isooctane flames, a detailed model describing high-temperature oxidation of both two fuels was obtained by merging the MTHF model from Bruycker et al. [21] and isooctane model from Chaos et al. [27]. The merged model consisting of 176 species and 1159 reactions performs well in predictions on ignition delay times and flame species profiles, and shows good predictions on MTHF laminar burning velocities under different conditions. The laminar burning velocities of MTHF/isooctane mixtures at atmosphere pressure are also predicted well by this model, suggesting it could be helpful to investigate the influence of the potential interaction between two fuels. This model for MTHF/isooctane mixtures could be found at Supplementary Material.

To calculate the laminar burning velocities under different conditions, numerical simulation was conducted using PREMIX code [31,32]. Grid-independent solutions were found with GRAD and CURV set as 0.01. Soret effect was taken into account and mixture-averaged transport was adopted.

3. Result and discussion

3.1. Experiment validation

The total experimental uncertainty of the laminar burning velocity measured by present facility is evaluated to be within ± 4.5 cm/s in previous studies [33,34]. In order to validate the experiment and the data processing method, the laminar burning velocities of isooctane in 373 K, 1 bar are compared against literature data [35–38] where spherical propagation flames were also researched under similar conditions, as shown in Fig. 1. Considering the differences in methodology used to extract the unstretched flame speed and the experimental uncertainty, the data in this work shows good consistency. There are more cases of validations in the Supplementary Material.

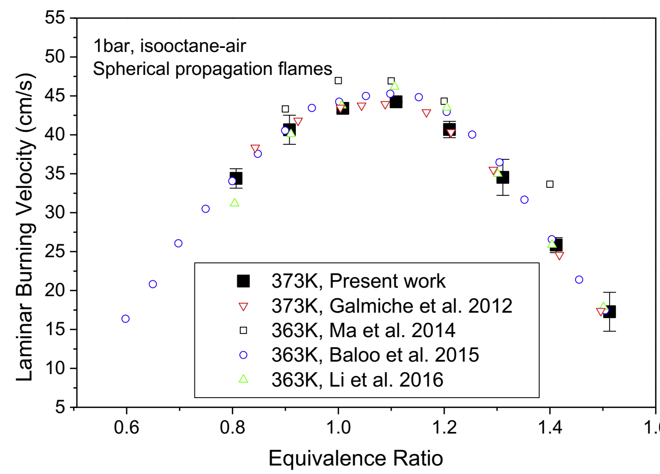

Fig. 1. Comparison of laminar burning velocities of isooctane/air at 1 bar. All the literature data is from spherical propagation flames: Baloo et al. [35] (T_u = 363 K), Galmiche et al. [36] (T_u = 373 K), Li et al. [37] (T_u = 363 K) and Ma et al. [38] (T_u = 363 K).

3.2. Laminar burning velocities of MTHF

Figure 2 shows the laminar burning velocities of MTHF–air mixtures under different conditions. The maximum laminar burning velocity occurs at a slightly fuel rich region near the equivalence ratio of 1.1. Adiabatic flame temperature has long been recognized to have a remarkable effect on burning velocity due to its influence on chemical kinetic, species equilibrium and the diffusion of heat and radicals into flame front. Effected by the amount of heat release and the product dissociation, the peak of adiabatic flame temperature is at the fuel rich side and consequently the peak of laminar burning velocity is also at the fuel rich side for most fuels. Figure 2(a) illustrates the laminar burning velocities of MTHF–air mixtures at temperatures of 373 K, 423 K and 453 K under atmosphere pressure. The peak values obtained in this experiment are 52.6, 62.6, 74.0 cm/s at the equivalence ratio of 1.1 in 373 K, 423 K, 453 K, respectively. The laminar burning velocities increase distinctly with initial temperature at different equivalence ratios. Most of that should be attributed to adiabatic flame temperature, diffusion and density. The increase in initial temperature increases the adiabatic flame temperatures and the thermal diffusion and consequently the laminar burning velocities of the mixtures.

The laminar burning velocities of MTHF at 423 K and pressures of 1, 2 and 4 bar are shown in Fig. 2(b). The laminar burning velocity decreases with the increase in pressure. As the initial pressure increases, the thermal diffusivity of mixture is reduced and the three-body, inhibiting reactions are increasingly important and consequently the overall chemical kinetics and the flame propagation slow down [39]. The uncertainty increases with pressure for fuel rich mixtures due to the appearance of unstable flame. In order to investigate the chemical kinetics of MTHF laminar combustion, a detailed model for MTHF and isooctane oxidation was adopted to analyze the combustion process.

3.3. Chemical kinetics analysis for MTHF

3.3.1. Model integration and validation

A detailed chemical kinetic model describing pyrolysis and combustion chemistry of MTHF was developed by Bruycker et

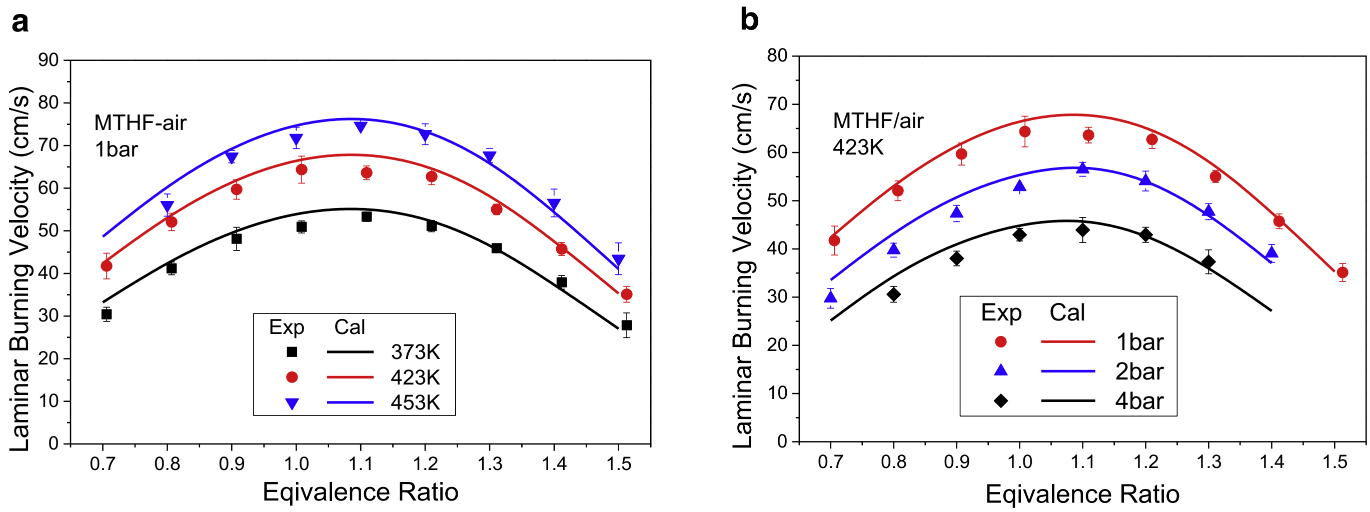

Fig. 2. Experimental and calculated laminar burning velocities of MTHF/air mixtures, Symbols for experimental data; Lines for calculations of the merged model.

al. [21] and validated against ignition delay times and speciation data of MTHF pyrolysis and oxidation. The fuel initial reactions and the decompositions of fuel-derived radicals were systematically studied through CBS-QB3 level theoretical calculations. In this study, to investigate the influence of the potential interaction between MTHF and isooctane, this MTHF mechanism [21] was merged with the isooctane model from Chaos et al. [27]. This isooctane model encompasses detailed high-temperature pathways for isooctane oxidation and pyrolysis and was validated against ignition delay times and laminar burning velocities. The base mechanism describing C1–C4 kinetics of this isooctane model is from Zhao et al. [40].

A MTHF/iso-octane model was assembled by merging the models of Bruycker et al. [21] and Chaos et al. [27]. While assembling the mechanism, the species and the reaction routes in the MTHF sub-mechanism from Bruycker et al. [21] was integrated into the isooctane model from Chao et al. [27]. Decomposition reactions from MTHF model [21] were added for some species formed in the MTHF sub-mechanism and for which no consumption routes are included in the isooctane base mechanism. When the consumption routes for some species in both the two models are inconsistent, the reactions in isooctane model were retained. Additional reactions from Tripathi et al. [18] and Bruycker et al. [21] were added for some important intermediates, such as 2,3-dihydro-5-methylfuran, 1-butene, acetaldehyde, acetyl radical, etc. Rates of several pressure dependent reactions were also updated.

Although the MTHF model from Bruycker et al. [21] slightly overestimates the laminar burning velocities of MTHF–air mixtures at high temperatures, as shown in previous work [21] and Supplementary Material, the simulations of the merged model exhibit improved consistency with the experiment data for most conditions, as shown in Fig. 2. Besides, this mechanism shows reasonable predictions on ignition delay and flame structures of MTHF [16,18,22], as shown in Supplementary Material. In this paper, the chemistry of MTHF oxidation in the flame will be analyzed using the merged mechanism.

3.3.2. Reaction pathway analysis for MTHF

Figure 3 illustrates the reaction pathway analysis of MTHF oxidation in the premixed flame at the initial temperature of 453 K, initial pressure of 1 bar and equivalence ratios of 0.8 and 1.3. There are several consumption pathways for MTHF, and the main consumption pathways under this condition are through H abstraction reactions to generate fuel radicals. The fuel radicals are unstable and generally undergo ring opening by C–C β scission. The intermediates from ring-opening reactions subsequently decompose into smaller fragments. As shown in Fig. 3, H abstraction reactions of MTHF at sites 2 and 5 are the main pathways for fuel consumption. They account for 68% of fuel consumption at 0.8 equivalence ratio and less for fuel rich combustion. The H abstraction reactions of MTHF at methyl group are less competitive. Instead of through ring opening, most of the tetrahydro-2-methyl-3-furanyl (Ccy(COCCCJ)) undergo CH_3 elimination producing 2, 3-dihydrofuran (cy(OC*CCC)) which is consumed via more complex processes than other intermediates. The branch ratios for most fuel consumption reactions don't present significant differences at different equivalence ratios. However, 15% of the fuel is consumed by C–C scission of the methyl side-group at the equivalence ratio of 1.3, and the proportion would reduce to close to 3% for fuel lean conditions.

Figure 4 shows several intermediate products of large mole fractions during the combustion at equivalence ratio of 1.3. Most of ethylene are from β-scission of resulting radicals from tetrahydro-2-methyl-2-furanyl (Ccy(CJOCCC)) scission and are consumed through H abstraction by H atom to generate vinyl radical (C_2H_3) which will generally go through H elimination to acetylene (C_2H_2). The resulting radicals formed from tetrahydro-2-methyl-5-furanyl (Ccy(COCJCC)) will mostly undergo β-scission reactions leading to propylene (C_3H_6) and vinyl-oxy radical (CH_2CHO). This is one of the main source of propylene, and the other is through recombination of H and allyl radical (C_3H_5-A). The consumption of propylene is mainly by H abstraction reactions from allylic site or replacement reaction with H atom to generate ethylene and methyl radical. Acetaldehyde (CH_3CHO) and symmetric allyl radical (C_3H_5-A) are mainly generated by two consecutive β-scission of tetrahydro-2-methyl-4-furanyl (Ccy(COCCJC)). Formaldehyde (CH_2O) also has a high concentration, most of which is produced by $CH_3 + O = CH_2O + H$ and scission of radicals from tetrahydro-2-methyl-1-furanyl (CJcy(COCCC)). The consumption of formaldehyde is mainly through H abstraction to formyl radical (HCO) in the flame.

3.3.3. Sensitivity analysis for MTHF

Sensitivity analysis is performed at 453 K, 1 bar and the equivalence ratios of 0.8, 1, 1.3 to identify the most influential reactions to laminar burning velocity, as shown in Fig. 5. The increase in the rate of the reaction with positive sensitivity will enhance the laminar burning velocity. The ability to accurately predict

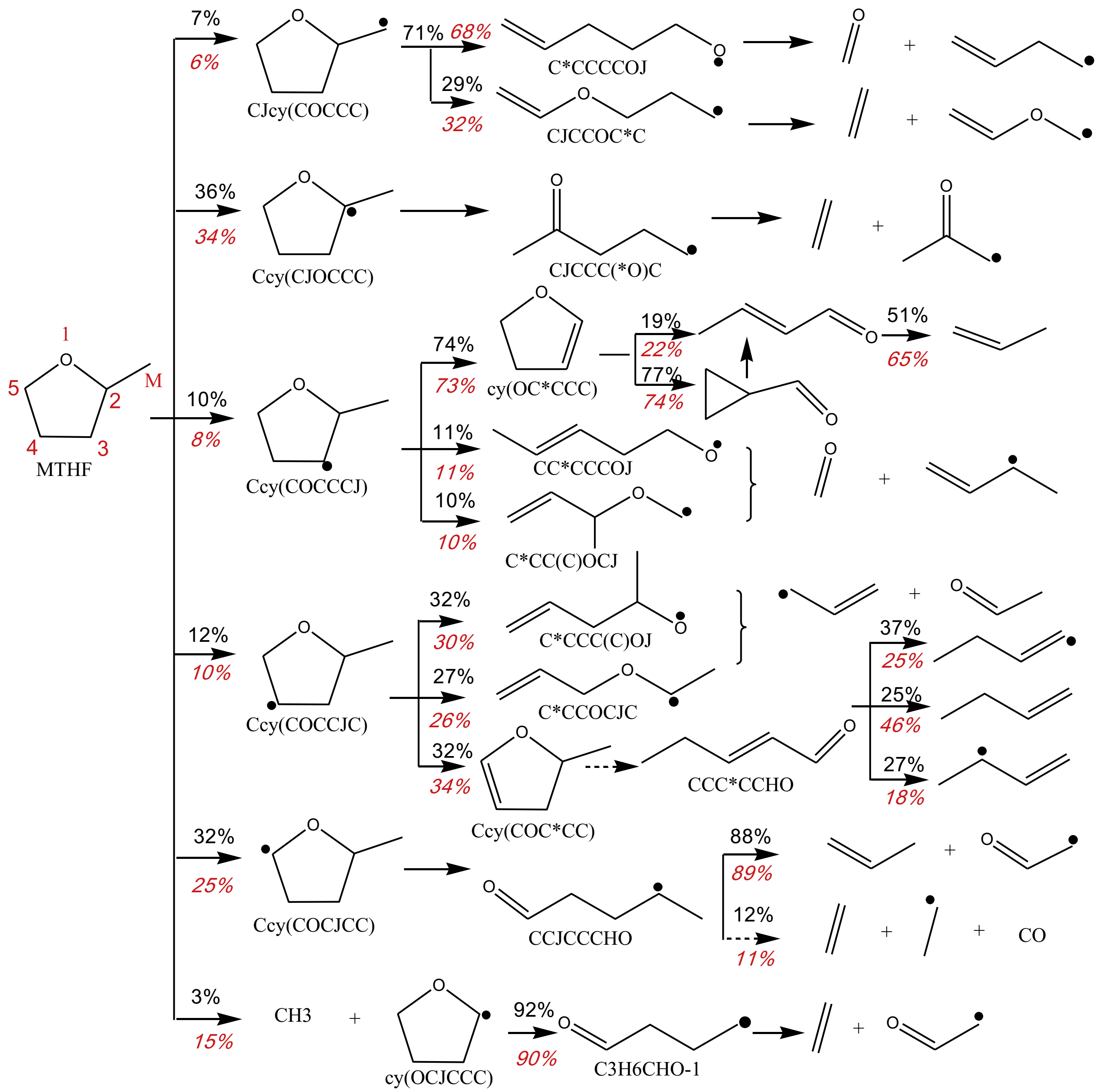

Fig. 3. The reaction path analysis for MTHF at 453 K, 1 bar. Black font for $\phi = 0.8$, red italic font for $\phi = 1.3$.

laminar-burning velocities, however, does not only depend on kinetic information involving the fuel itself. Reactions of small species and important intermediates have great influence on laminar burning velocity. The sensitivities of fuel initial reactions are relatively weak. Only one fuel initial reaction, the reaction of C–C scission of the methyl group, is in the top 30 sensitive reactions. The increase in the rate of this reaction will evidently enhance the laminar burning velocity at fuel rich conditions. The H abstraction reaction of propene to generate allyl radical (C_3H_5-A) and the recombination reaction of H and allyl radical to generate propylene show negative sensitivities because these two reactions consume a great amount of reactive radicals, thereby would reduce the laminar burning velocity.

3.4. Laminar burning velocities of MTHF/Isooctane

3.4.1. Laminar burning velocity

Figure 6 shows the laminar burning velocities of blended fuels at initial temperature of 373 K and 423 K. The laminar burning velocities increase with initial temperature. The adiabatic flame temperature and the thermal diffusion increase with initial temperature and exhibited positive influence on the laminar burning velocities. The laminar burning velocities reach the peak value at the equivalence ratio near 1.1. The simulation shows reasonable agreement with experimental data under most conditions, as shown in Fig. 6. For higher initial temperatures and pressures, the prediction discrepancies for isooctane and blended fuels increased with equivalence ratio. Most probably the chemistry

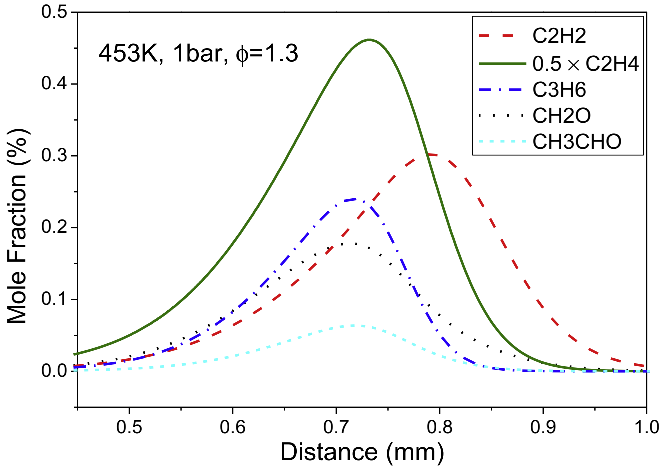

Fig. 4. Mole fractions of important intermediates at 453 K, 1 bar. Note that the mole fraction of C_2H_4 is divided by 2.

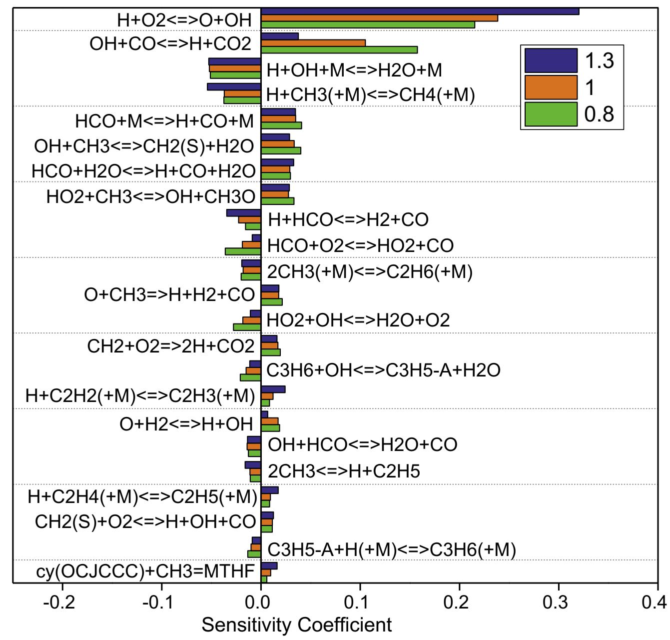

Fig. 5. The sensitivity analysis for MTHF at 453 K, 1 bar.

model should be further extended toward heavier hydrocarbons for rich isooctane flames.

The laminar burning velocities of MTHF are the highest at all the equivalence ratios while that of isooctane are the lowest among the five fuels. The laminar burning velocities increase with MTHF blending ratio. For MTHF mole fraction of 20%, the corresponding volume fraction is 13.17%, and its laminar burning velocity is quite close with that of the isooctane. For MTHF mole fraction of 60%, the corresponding volume fraction is 47.64% and the laminar burning velocity is generally at the average position of that of MTHF and isooctane.

3.4.2. Factors affecting laminar burning velocities between different blending ratios

In addition to the chemical factor, the laminar burning velocity could also be affected by thermal and diffusion factors, which could be characterized by adiabatic flame temperature, thermal diffusivity and effective Lewis number. The increase in adiabatic flame temperature and thermal diffusivity can increase the laminar burning velocity. The effective Lewis number is a weighted average of the individual Lewis numbers of the reactant. For very lean/rich mixtures, the effective Lewis number is practically the Lewis number of the fuel/oxidizer [30], representing the ratios of the thermal diffusivity of the mixture to the mass diffusivities of the deficient reactant. However, the increase in mass diffusion of the deficient reactant leads to concentration decrease in unburned mixture and the slowdown of laminar burning velocity. The laminar burning velocity could be affected by MTHF blending through these parameters.

The adiabatic flame temperatures and thermal diffusivities are very close at different blending ratios, as shown in Fig. 7. Besides, in view of the similar thermal diffusivity among the blended fuels, the decrease in effective Lewis number with increasing blending ratio indicates the rise of mass diffusivity of the deficient reactant, resulting in negative effect of MTHF blending on laminar burning velocity. However, the laminar burning velocity of MTHF is the highest among the five fuels. So, the increase in laminar burning velocity with increasing MTHF blending ratio should mostly attribute to the chemical factors.

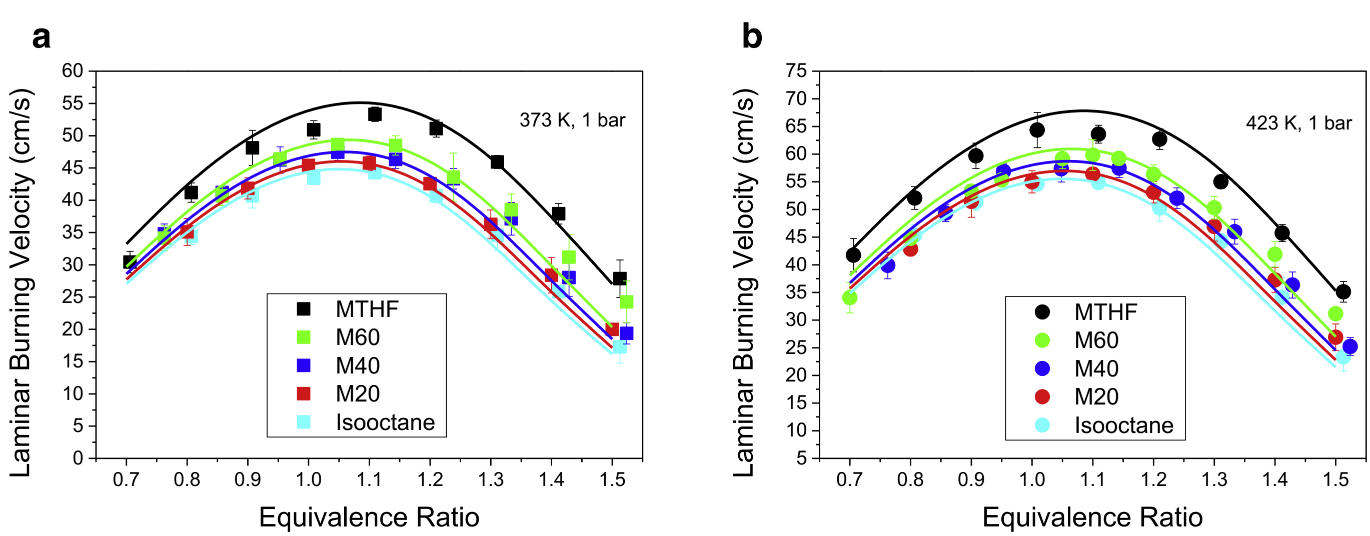

Fig. 6. The laminar burning velocity of MTHF/isooctane blends at 373 K (a) and 423 K (b). Symbols for experimental data; Lines for simulation.

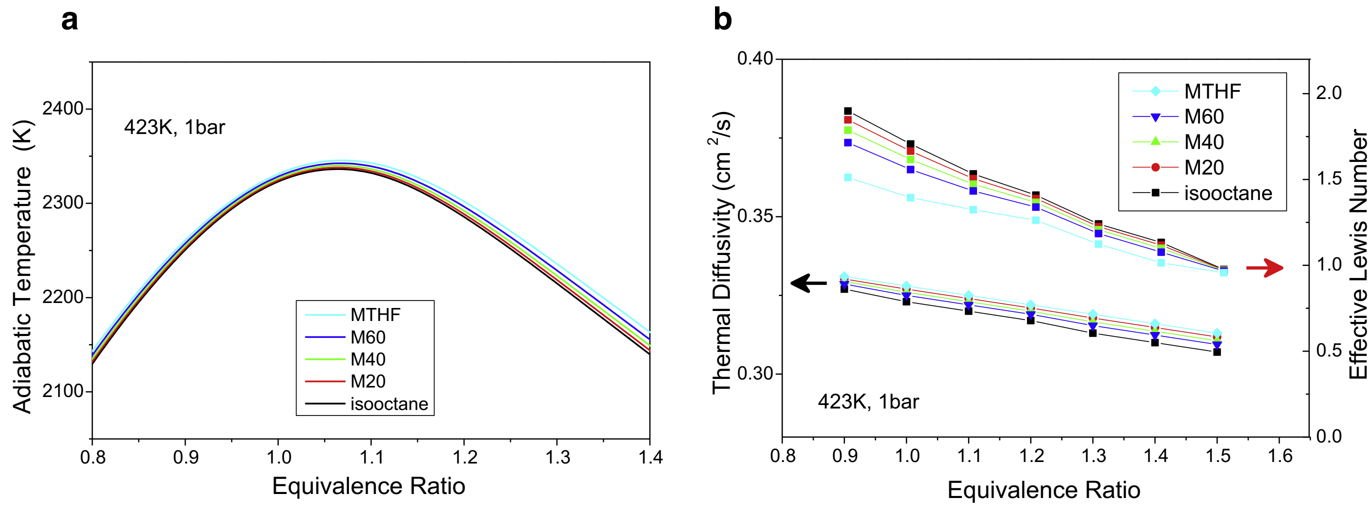

Fig. 7. The adiabatic flame temperature (a), the effective Lewis number and the thermal diffusivity (b) of blend fuels at 423 K, 1 bar.

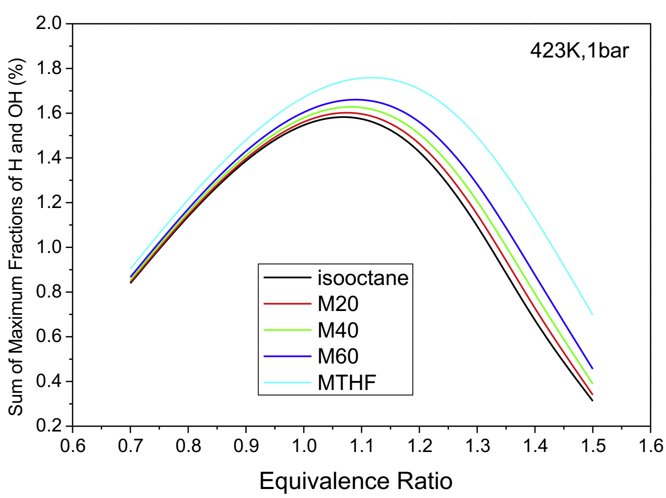

Fig. 8. The sum of the maximum fractions of H and OH in flames at 423 K, 1 bar.

3.4.3. Chemical kinetic effects

The structures and consumption pathways of fuel molecules and intermediate products are the main reasons for the difference in laminar burning velocities between the two fuels. To further analyze this chemical kinetic effect, several analyses were adopted for MTHF/isooctane blends.

Reactive radicals play an important role in initial reactions and chain branching reactions, thus could strongly affect the overall reaction rate and combustion intensity. Figure 8 shows the sum of the maximum concentrations of H and OH at different equivalence ratios. MTHF combustion produces reactive radical pools of higher concentration than that of isooctane. In the case of MTHF addition into isooctane, H and OH concentration would increase and the overall reaction rate and laminar burning velocities would be promoted. Moreover, the strong diffusion capacity of these small radicals also has a positive effect. The disparities of radical concentrations between different blending ratios increase for fuel rich mixtures, indicating that the effect of reactive radicals enhanced.

In the pathway analysis for isooctane and MTHF, the branching ratio of fuel decomposition decreased as the increase in MTHF blending ratios, while the proportion of H abstraction by H, OH and O increased, as illustrated in Fig. 9. This is due to the increases in the amounts of these reactive radicals at high MTHF blending ratios.

As shown in Fig. 10, the H abstraction reaction of propylene to generate resonantly stabilized allyl radical (C_3H_5-A) and the recombination reaction of H and allyl radical to generate propylene show negative sensitivities. These two reactions consume a great amount of reactive radicals, thereby would reduce the laminar burning velocity. As a consequence, although the competitive propylene consumption reaction, $C_3H_6 + H = C_2H_4 + CH_3$, could produce methyl radical (CH_3), the sensitivity is positive. In the pathway analysis for isooctane, there are many ways to produce propylene, such as the demethylation and replacement reactions of isobutene (IC_4H_8), isopropyl radical (IC_3H_7) and isobutyl radical (IC_4H_9), the decomposition of C6–C8 intermediates, etc. The pathways to generate propylene are less for MTHF than that for isooctane, inducing the decrease in production rate and concentration of propylene as MTHF blending ratio increase. As shown in Fig. 11(a), the proportion of propylene in MTHF flame is approximately half of that in isooctane flames, which is one of the reasons why the laminar combustion velocity of MTHF is higher than that of isooctane.

There is also considerably more ethylene (C_2H_4) in the flames of MTHF than that of isooctane, because extensive amount of ethylene is produced through β scission of ring opening products of tetrahydro-2-methyl-2-furanyl (Ccy(CJOCCC)) in MTHF flame. However, this reaction does not show strong sensitivity. Methyl radical (CH_3) has a negative effect on the combustion velocity, because it will undergo the chain termination reactions, $2CH_3 = C_2H_6$, $H + CH_3 = CH_4$, to generate stable intermediates. These two reactions and the CH_3 generation reaction $2CH_3 = H + C_2H_5$, have strong negative sensitivities, as shown in Fig. 11(b). Isooctane molecular has more methyl group, but the content of methyl radical in isooctane flame is quite close to that in MTHF flame, as shown in Fig. 11(b).

The common intermediates of MTHF and isooctane are mostly the molecules of C0–C3, because the isooctane intermediates mostly have branched structure due to the structure of isooctane. Species with branched structure have been shown to have low reactivity, such as the laminar flame speeds of isomers of butane [41], butene [42], butanol [43], octane [44,45].

3.4.4. Correlation of laminar burning velocity

In order to further reveal the relationship between the laminar burning velocity and the blending ratio, the experimental data

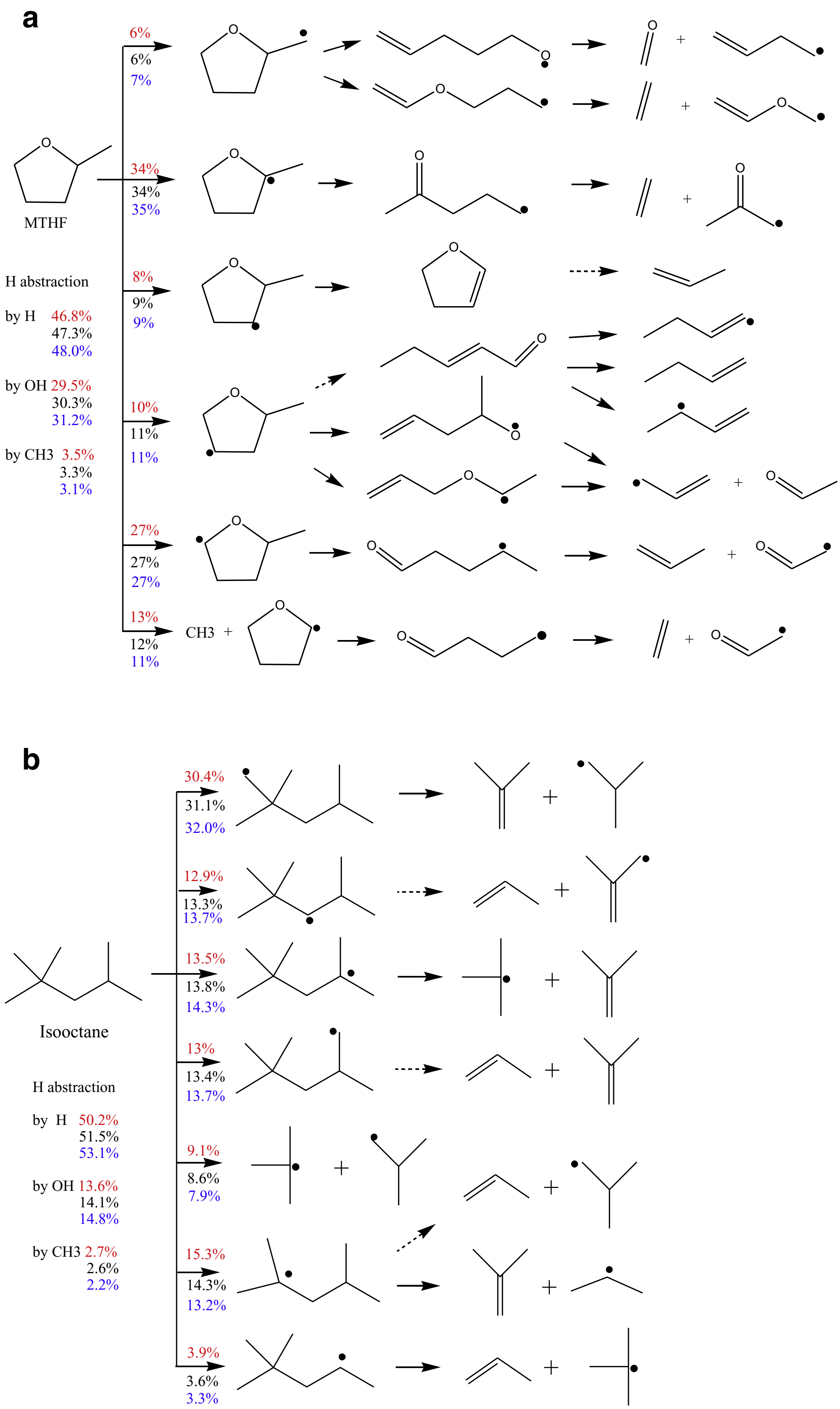

Fig. 9. The reaction pathway of MTHF (a) and isooctane (b) in M20 (red font), M40 (black font) and M60 (blue font) flames at 373 K, 1 bar, with equivalence ratio of 1.1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

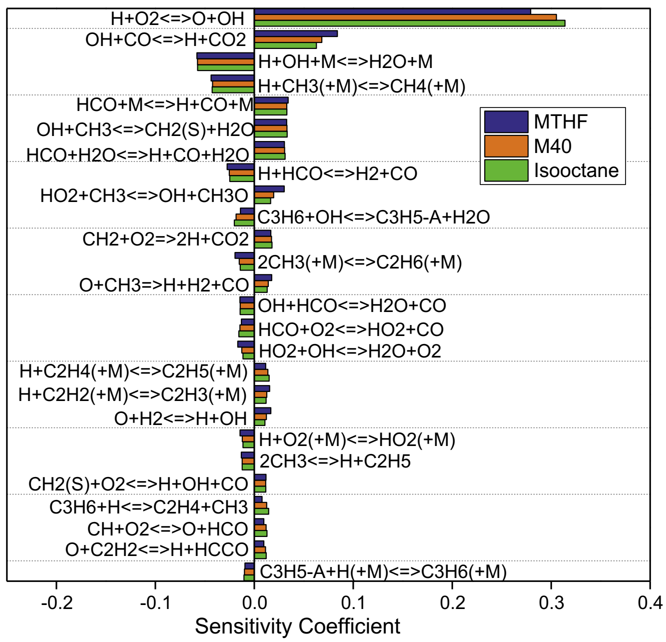

Fig. 10. The reaction sensitivities in MTHF, M40 and isooctane flames at 373 K, 1 bar, with equivalence ratio of 1.1.

were correlated according to blending ratio, temperature, pressure and equivalence ratio. Hirasawa et al. [46] explored the mixing rule of laminar burning velocities for binary fuel blends of ethylene, n-butane, and toluene. Ji et al. [47] analyzed the adiabatic flame temperature and laminar burning velocities of binary fuel blends and the influence of kinetic couplings. Broustail et al. [48] investigated the laminar burning velocities of butanol/isooctane and ethanol/isooctane blends under different pressures, and utilized the mixing rule suitable for binary fuel blends which have similar adiabatic flame temperatures at different blending ratios. Based on these studies above, the mixing rule in exponential form are utilized for present work:

$$S_m = \exp\left[\sum_{i=1}^{2} \chi_i \ln(S_i)\right] \tag{9}$$

$$\chi_i = \frac{X_i N_i T_{ad,i}}{N_m T_{ad,m}} \tag{10}$$

Table 2
The fitting parameters in Eqs. (9)–(14).

	MTHF	isooctane
A	66.07	55.85
B	19.60	10.46
C	−211.35	−203.95
D	−36.81	−15.51
E	248.20	227.84
F	1.79	1.79
G	0.25	1.83
H	−0.30	30.44
I	0.09	0.14
ϕ	1.05	1.05

where m refers to the blend fuels, i refers to MTHF or isooctane, S is the laminar burning velocity, X is the blending ratio in mole, N is the total mole of products of per mole of fuel, Tad is the adiabatic flame temperature of the mixture, χ represents the weight of pure fuel to dominate the combustion process of binary fuel, and $\sum_{i=1}^{2} \chi_i = 1$ [47], S_i is the laminar burning velocity of pure fuel and could be correlated by temperature and pressure [49]:

$$S_i = S_{i,ref}\left(\frac{T}{T_{ref}}\right)^{\alpha_i}\left(\frac{P}{P_{ref}}\right)^{\beta_i} \tag{11}$$

$$S_{i,ref} = A_i + B_i(\phi - \phi_i) + C_i(\phi - \phi_i)^2 + D_i(\phi - \phi_i)^3 + E_i(\phi - \phi_i)^4 \tag{12}$$

$$\alpha_i = F_i + G_i(\phi - \phi_i) \tag{13}$$

$$\beta_i = H_i + I_i(\phi - \phi_i) \tag{14}$$

where A_i–I_i and ϕ_i are fitting parameters, T and P are temperature and pressure, ref is the reference condition of 423 K, 1 bar.

Least squares method was used in correlation for experiment data. Table 2 depicted the fitting parameters. The correlations capture the experiment data fairly well, as shown in Figs. 12 and 13. The average error is 2.3% and the maximum error is within 3 cm/s.

As shown in Fig. 14, the laminar burning velocities increase with MTHF blending ratio at different equivalence ratios. The fact that this mixing rule based on thermodynamic parameters can predict the flame speeds well suggests that the chemical kinetic coupling between isooctane and MTHF, if any, is of minimal importance to the flame speed at this range of temperature and pressure.

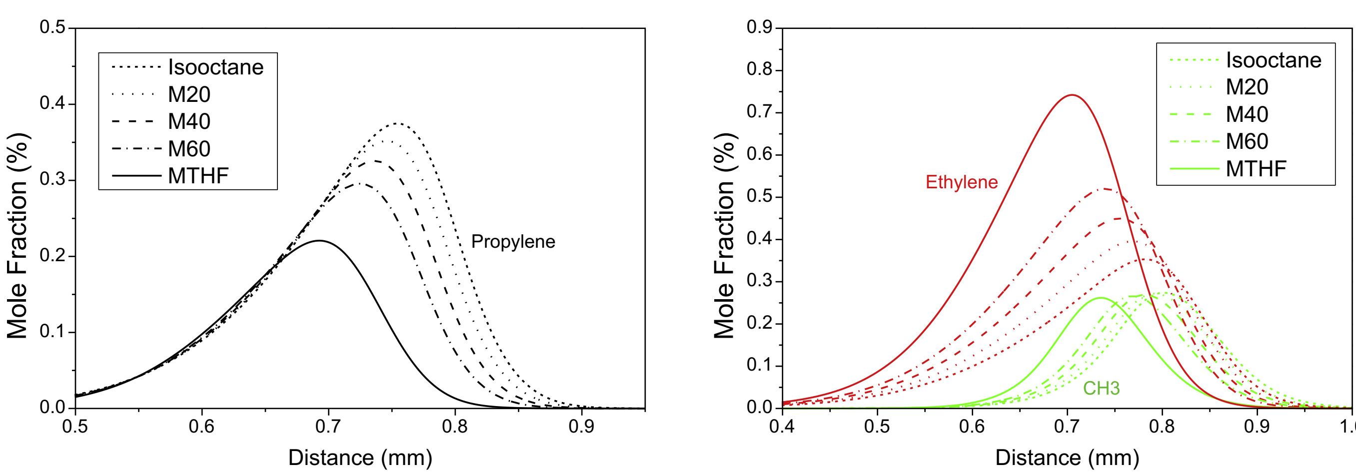

Fig. 11. The mole fraction of propylene (a), ethylene and methyl radical (b) in MTHF/ isooctane flames at 373 K, 1 bar, with equivalence ratio of 1.1.

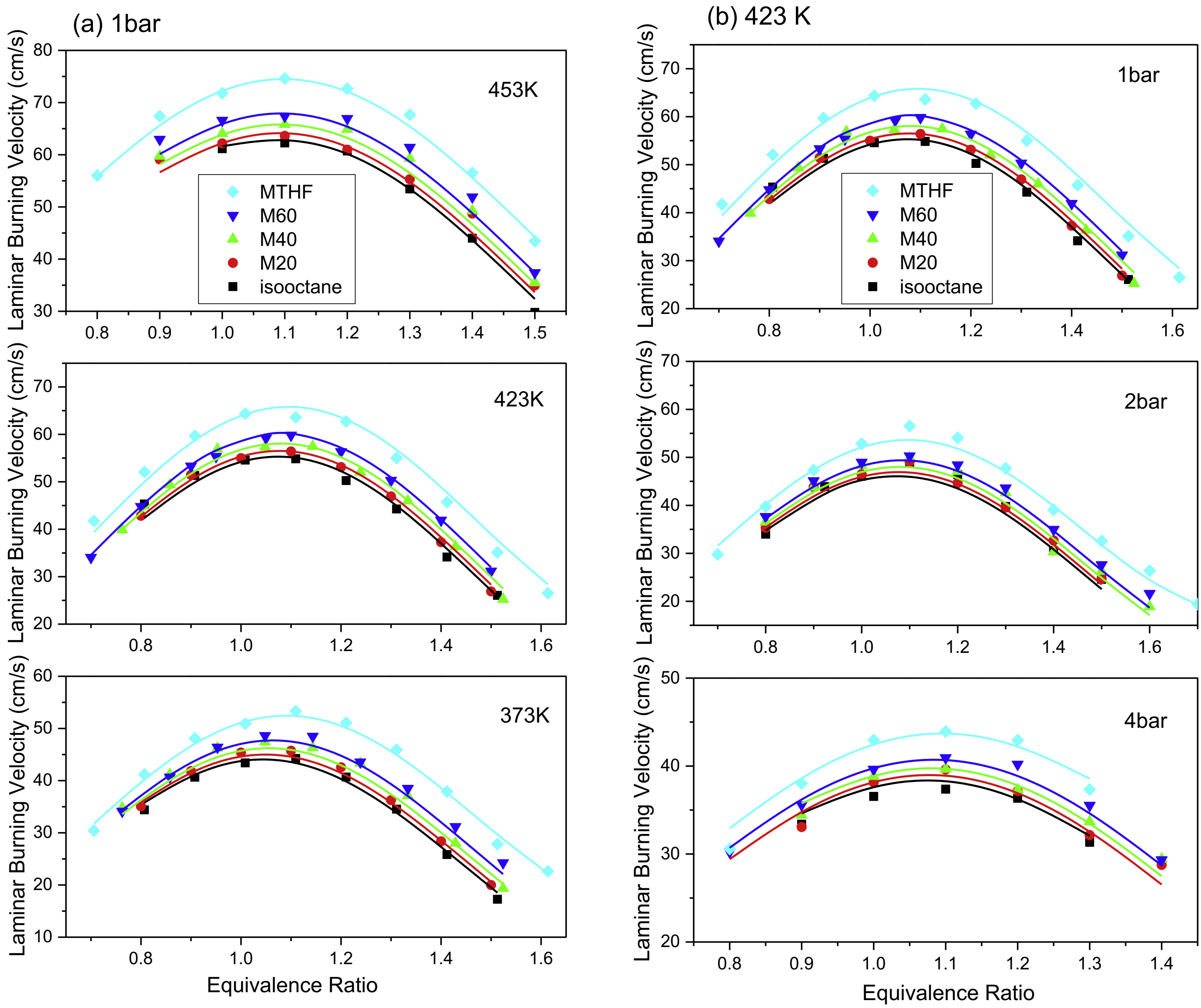

Fig. 12. the laminar burning velocity of MTHF/Isooctane at different temperatures at 1 bar (a), and at different pressures at 423 K (b). Symbols for experimental data; Lines for fitting correlations of Eqs. (9)–(14).

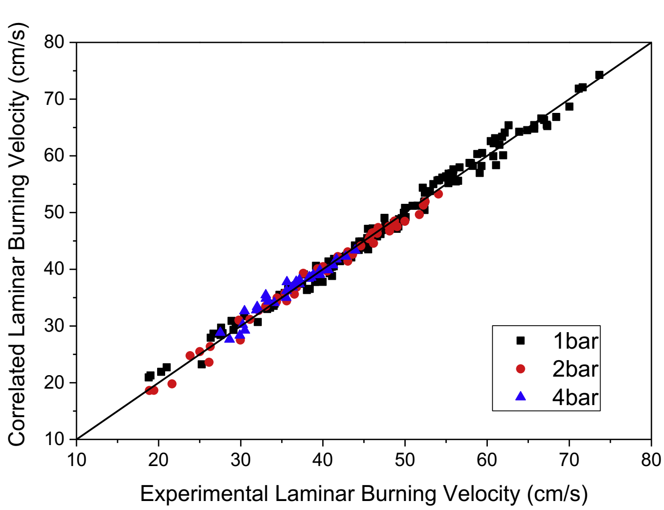

Fig. 13. The experimental and correlated laminar burning velocity.

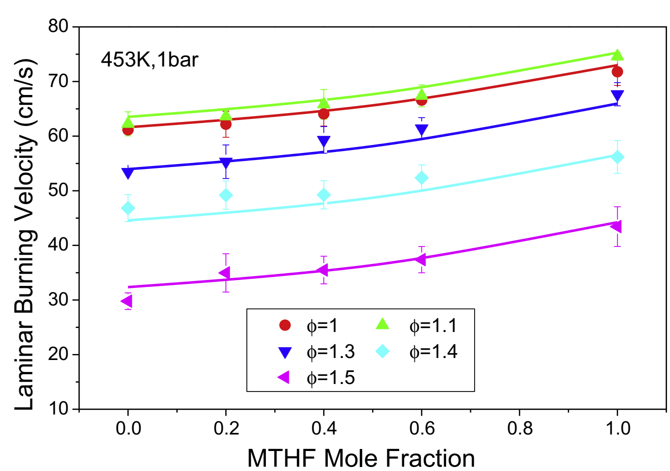

Fig. 14. The laminar burning velocities of different MTHF mole fractions at 453 K, 1 bar. Symbols for experimental data; Lines for fitting correlations of Eqs. (9)–(14).

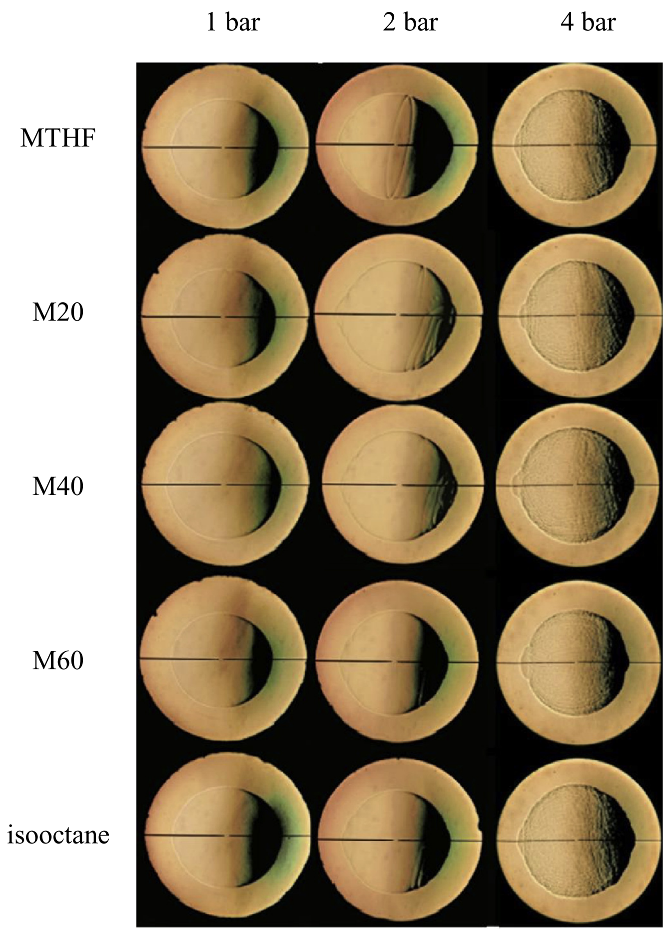

Fig. 15. The flames with radius of about 24 mm at equivalence ratio of 1.3 and different pressures. Cellular structures are observed in images at 4 bar.

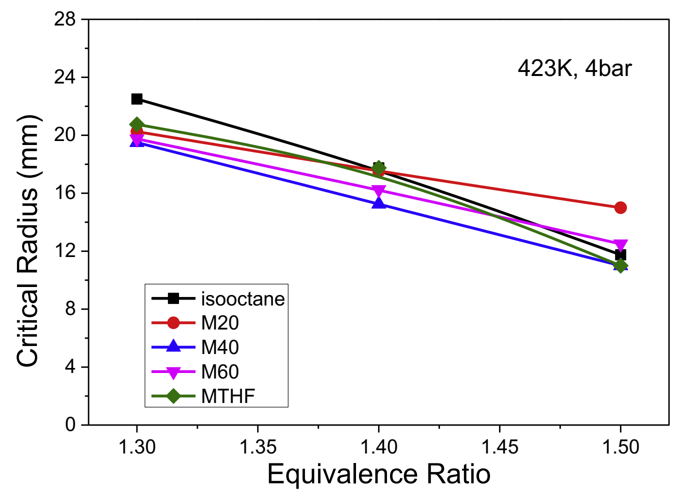

Fig. 16. The critical radius at 423 K, 4 bar.

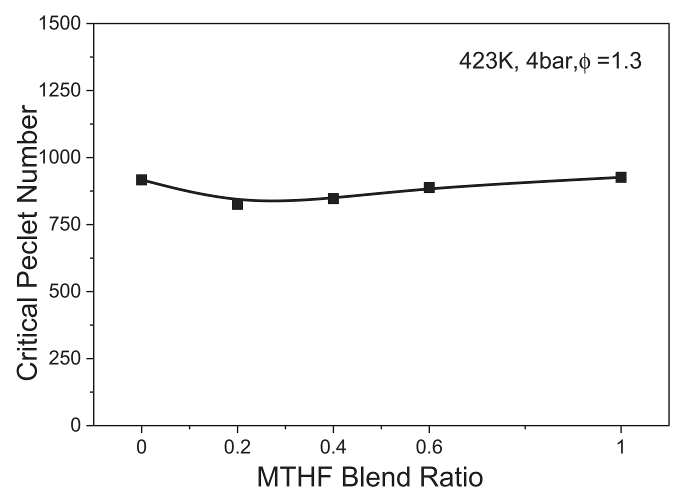

Fig. 17. The critical Peclet number.

3.5. The flame instability of MTHF/isooctane

3.5.1. Flame morphology and the critical radius

Figure 15 shows the schlieren pictures of several fuels at the temperature of 423 K, pressures of 1, 2 and 4 bar, equivalence ratio of 1.3. The flame surface keeps smooth at 1 bar for all blending ratios and some cracks appear on the photographs of 2 bar. However, the cracks do not further develop as flame propagates. At 4 bar, as the development of the flame, the instability increases and cellular structures are observed. Starting from a certain radius, the cracks formed at the beginning are increasingly deep and start to branch, then the cells spontaneously appear over the entire flame surface. The flame propagation speed increased rapidly at the onset of unstable flame. The difference of the flame morphology is not distinct among different blending ratios, as shown in Fig. 15.

The critical radius is the radius at which the stable flame surface transferred to unstable, and is defined in this study as the radius when the cell structures form in large quantity. Figure 16 shows that the critical radius decrease linearly with increasing equivalence ratio, indicating the enhancement of flame instability. The critical Peclet number is a dimensionless number defined as the ratio of critical radius to flame thickness. As shown in Figs. 16 and 17, the critical radius and the critical Peclet numbers are similar at different blending ratios, reflecting the impact of blending ratio on the flame instability is not remarkable.

Flame instability is mainly influenced by diffusive-thermal factor and hydrodynamic factor [30]. Several parameters evaluating the two factors, such as Markstein length, density ratio and the flame thickness, are discussed to further understand the flame instability at different blending ratios.

3.5.2. Markstein length

Markstein length could quantify the response of flame propagation to the stretch rate and is generally used to evaluate the diffusion-thermal instability. Small Markstein length is the indication of both less stable flame surface and small or positive impact of flame stretch rate on burning velocity.

From Fig. 18(a) and (b), the Markstein length are similar at different temperatures, consisting with the laminar combustion study of MTHF by Jiang et al. [24]. In general, the Markstein length tends to increase monotonically with equivalence ratio for light fuel–air mixtures and tends to decrease for the heavy hydrocarbon–air mixtures. MTHF and isooctane are heavy hydrocarbons and less diffusive than air. As a consequence, the Markstein length gradually decreases with the increase in equivalence ratio for MTHF and isooctane. The Markstein lengths of several fuels decrease evidently with the rise in initial pressure. At 4 bar, the Markstein lengths are close to zero, even for small equivalence ratios, suggesting the flame is less stable at high pressure. The Markstein lengths are similar for different blending ratios, reflecting the similar flame instability. The variations of Markstein length with blend-

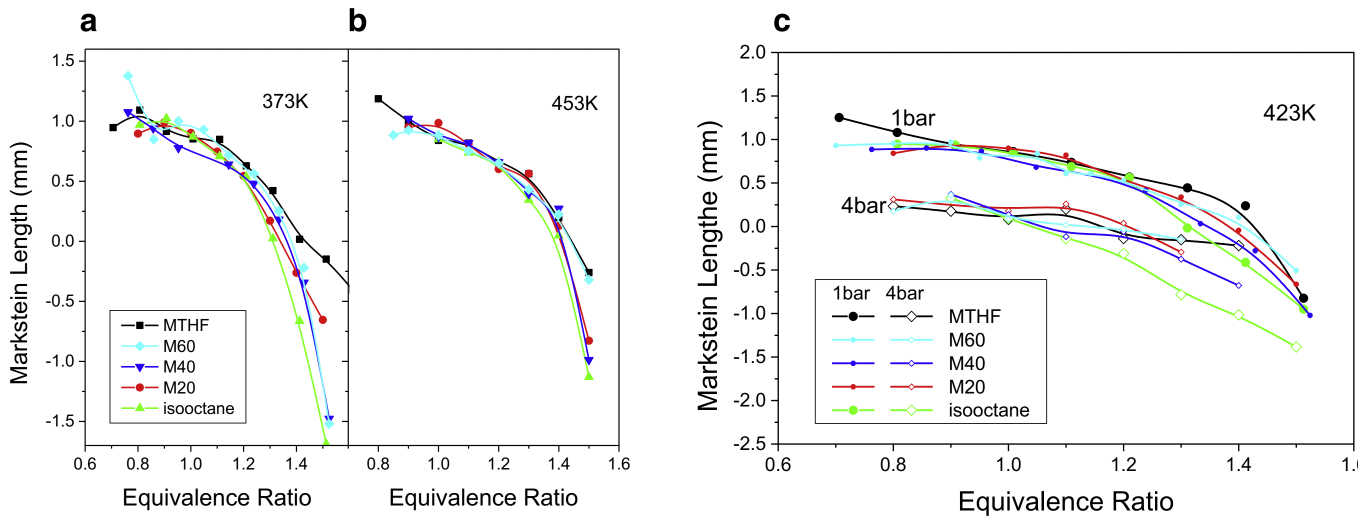

Fig. 18. The Markstein length of blended fuels.

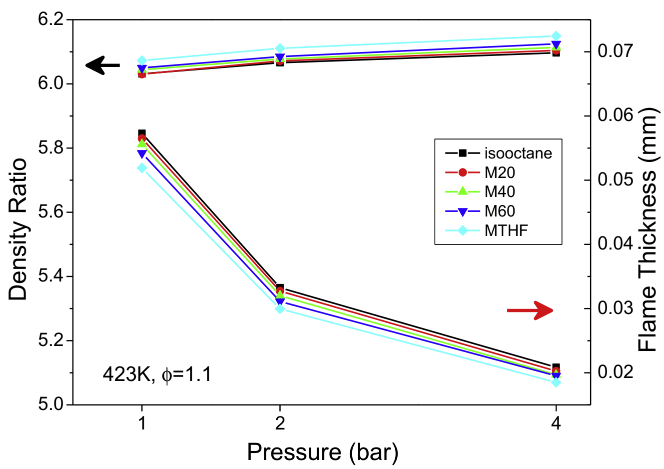

Fig. 19. The density ratio and flame thickness versus pressure.

ing ratio, pressure and equivalence ratio are consistent with that of the radius of the onset of cellular flame in experiments.

3.5.3. Flame thickness and density ratio

The hydrodynamic instability is due to the thermal expansion between the burned the unburned mixture and always exists during the flame propagation. In this study, the flame thickness is identified as the ratio of thermal diffusivity to laminar burning velocities, and the density ratio is determined as the ratio of the unburned densities to the burned. The reduction in the density ratio and the increase in the flame thickness could reflect the reduction in hydrodynamic instability. As shown in Fig. 19, the density ratios and flame thicknesses are close among different blending ratios while the flame thicknesses distinctively decrease with the rise of pressure. It can be concluded that the hydrodynamic instability is close for different MTHF blending ratios and significantly enhanced by pressure rise, consisting with Markstein length and critical radius.

4. Conclusions

In this study, the spherically propagating flames of the mixtures of MTHF/isooctane–air were explored in a constant volume chamber using high-speed photography technique. The laminar burning velocities of blend fuels were determined through nonlinear method and correlated by mixing rules as a function of initial temperature, pressure, equivalence ratio, and blending ratio. To investigate the chemical oxidation of MTHF/isooctane blends, a detailed model was established by integrating MTHF sub model into isooctane model. The merged model presents improved predictions on MTHF laminar burning velocities and has been validated against flame structure and ignition delay times. The chemical kinetics of MTHF oxidation was analyzed. The effect of blending ratio and other parameters on laminar burning velocity as well as the flame instability were discussed. Several conclusions could be collected.

The laminar burning velocity of blend fuels reaches the peak at equivalence ratio near 1.1. The laminar flame velocity was observed to increase with increase in MTHF blending ratio and initial temperature while the flame velocity decreases with increase in pressure. There are distinctive disparities between laminar burning velocities of MTHF and isooctane. The laminar burning velocity of M20 is close with that of the isooctane. The laminar burning velocity of M60 is generally at the average position of that of MTHF and isooctane.

Kinetic analysis indicates the primary pathway for MTHF consumption is through H abstraction reactions at 2 and 5 sites. The H abstraction reactions at methyl group are less competitive. The branch ratios for most fuel consumption reactions don't present significant differences at fuel rich and lean conditions, except for the C–C scission at methyl group. Through β scission of ring opening products of fuel radicals, unsaturated hydrocarbons (ethylene, propylene) and aldehydes (formaldehyde, acetaldehyde) are produced in high concentration and are the main stable intermediates of MTHF combustion. Sensitivity analysis shows that reactions of small species have great influence on laminar burning velocity, but the sensitivities of initial reactions are relatively weak. Some main intermediates like propylene and ethylene, also show strong influence.

The effect of MTHF blending on laminar burning velocity should attribute to the chemical factors, rather than thermal or diffusion factors. The structures and consumption pathways of fuel molecules and intermediate products are the main reasons for the difference in laminar burning velocities between the two fuels. The common intermediates of MTHF and isooctane are mostly the species of C0–C3 due to the difference in fuel molecular structures. Some of these species have great influence on laminar burning velocity. The discrepancy in the concentrations of H, OH, and propylene between the flames of MTHF and isooctane reflects the positive chemical effect on laminar burning velocity as MTHF blended, especially for fuel rich mixtures.

As exhibited by Markstain length and flame thickness, the flame instabilities for different blending ratios increase with pressure and equivalence ratio. However, the Markstein length and flame thickness are similar for different blending ratios, reflecting the similar flame instability. The critical radius and critical Peclet number do not show distinct effect of MTHF blending ratio on flame instability either.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2017.12.028.

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